Exercise J - A

1. Consider the system of equations

$x - 2y + 3z = -1, -x + y - 2z = k, x - 3y + 4z = 1$

STATEMENT-1 : The system of equations has no solution for $k \neq 3$.

and

STATEMENT-2 : The determinant $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$, for $k \neq 3$. **[JEE-2008]**

(A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1

(B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

समीकरणों के निकाय

$x - 2y + 3z = -1$

$-x + y - 2z = k$

$x - 3y + 4z = 1$

कथन-1 : $k \neq 3$ के लिये समीकरणों के निकाय का कोई हल नहीं है।

तथा

कथन-2 : $k \neq 3$ के लिये सारणिक $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$ है।

(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है

(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण नही है

(C) कथन–1 सत्य है, कथन–2 असत्य है

(D) कथन–1 असत्य है, कथन– 2 सत्य है

Ans. A

Sol. $x - 2y + 3z = -1, -x + y - 2z = k$

and $x - 3y + 4z = 1$

$\Delta = \begin{vmatrix} 1 & -2 & 3 \\ -1 & 1 & -2 \\ 1 & -3 & 4 \end{vmatrix} = 0$ and $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} = 3 - k$

Hence if $k = 3$ then system will have infinite solutions and $k \neq 3$ then system will have no solution. so S(I) and S(II) both are true & (II) is correct explaination for (I).

2. The number of all possinle values of θ, where $0 < \theta < \pi$, for which the system of equations

$(y + z)\cos 3\theta = (xyz)\sin 3\theta$

$$x \sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$$

$(xyz)\sin 3\theta = (y + 2z)\cos 3\theta + y \sin 3\theta$

have a solution $(x_0, y_0 z_0)$ with $y_0 z_0 \neq 0$, is. **[JEE-2010]**

θ के उन सभी सम्भावित मानों की संख्या ज्ञात कीजिये, जिनके लिए समीकरण के निकाय

$(y + z)\cos 3\theta = (xyz)\sin 3\theta$

$$x\sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$$

$(xyz)\sin 3\theta = (y + 2z)\cos 3\theta + y \sin 3\theta$ का एक हल $(x_0, y_0 z_0)$ है, जहाँ $y_0 z_0 \neq 0$ तथा $0 < \theta < \pi$, है।

[JEE-2010]

Ans. 3

Sol. $(y + z)\cos 3\theta = (xyz)\sin 3\theta$ ...(i)

$$x \sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z} \quad ...(ii)$$

$(xyz)\sin 3\theta = (y + 2z)\cos 3\theta + y\sin 3\theta$ ...(iii)

where $yz \neq 0$ and $0 < \theta < \pi$

from (i) & (iii)

$(y + z)\cos 3\theta = (y + 2z)\cos 3\theta + y \sin 3\theta$

$\Rightarrow$ $z \cos 3\theta + y \sin 3\theta = 0$ ...(iv)

from eq^n (ii)

$2z \cos 3\theta + 2y \sin 3\theta = xyz \sin 3\theta$ ...(v)

from equation (iv) & (v)

$\Rightarrow$ $xyz \sin 3\theta = 0$

$\Rightarrow$ $x \sin 3\theta = 0$ as $yz \neq 0$

Possible cases are either $x = 0$ or $\sin 3\theta = 0$

Case (1) : if $x = 0$

$\Rightarrow$ $y + z = 0 \Rightarrow y = -z$

from eq^n (iv) $\cos 3\theta = \sin 3\theta$

$$\Rightarrow \quad 3\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}$$

$$\Rightarrow \quad \theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}$$

Case (2) : if $\sin 3\theta = 0$

$\Rightarrow \quad \theta = \frac{\pi}{3}, \frac{2\pi}{3}$

But these values does not satisfy given equations.

Hence, total number of possible values of θ are 3.

3. Which of the following values of α satisy the equation $\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\,\alpha$?

[JEE-Adv.-2015]

α के निम्नलिखित मानों में कौन सा मान समीकरण $\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\,\alpha$ को संतुष्ट करता है?

[JEE-Adv.-2015]

(A) –4 (B) 9 (C) –9 (D) 4

Ans. BC

Sol. $R_3 \to R_3 - R_2, R_2 \to R_2 - R_1$

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ 3+2\alpha & 3+4\alpha & 3+6\alpha \\ 5+2\alpha & 5+4\alpha & 5+6\alpha \end{vmatrix} = -648\alpha$$

$R_3 \to R_3 - R_2$

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ 3+2\alpha & 3+4\alpha & 3+6\alpha \\ 2 & 2 & 2 \end{vmatrix} = -648\alpha$$

$C_3 \to C_3 - C_2, C_2 \to C_2 - C_1$

$$\begin{vmatrix} (1+\alpha)^2 & \alpha(2+3\alpha) & \alpha(2+5\alpha) \\ 3+2\alpha & 2\alpha & 2\alpha \\ 2 & 0 & 0 \end{vmatrix} = -648\alpha$$

$\Rightarrow \quad 2\alpha^2(2+3\alpha) - 2\alpha^2(2+5\alpha) = -324\alpha$

$\Rightarrow \quad -4\alpha^3 = -324\alpha \Rightarrow \quad \alpha = 0, \pm 9$

4. The total number of distinct $x \in \mathbb{R}$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is. **[JEE-Adv.-2016]**

ऐसे सभी भिन्न $x \in \mathbb{R}$, जिनके लिए $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ है, की कुल संख्या है। **[JEE-Adv.-2016]**

Ans. 2

Sol. $\begin{vmatrix} x & x^2 & 1 \\ 2x & 4x^2 & 1 \\ 3x & 9x^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 2x & 4x^2 & 8x^3 \\ 3x & 9x^2 & 27x^3 \end{vmatrix} = 10$

Using $\begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} = (a-b)(b-c)(c-a)$

Again $\begin{vmatrix} x & x^2 & 1 \\ 2x & 4x^2 & 1 \\ 3x & 9x^2 & 1 \end{vmatrix} + (x)(2x)(3x)\begin{vmatrix} 1 & x & x^2 \\ 1 & 2x & 4x^2 \\ 1 & 3x & 9x^2 \end{vmatrix} = 10$

$\Rightarrow (x-2x)(2x-3x)(3x-x) + 6x^3[(x-2x)(2x-3x)(3x-x)] = 10$

$\Rightarrow 2x^3(1+6x^3) = 10$ (solve by putting $x^3 = t$)

$\Rightarrow x = -1, \left(\frac{5}{6}\right)^{\frac{1}{3}}$

5. Let $a, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations

$ax + 2y = \lambda$

$3x - 2y = \mu$

Which of the following statement(s) is(are) correct ? **[JEE-Adv.-2016]**

(A) If $a = -3$, then the system has infinitely many solutions for all values of λ and μ

(B) If $a \neq -3$, then the system has a unique solution for all values of λ and μ

(C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$

(D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$

माना कि a, $\lambda, \mu \in \mathbb{R}$ हैं। इन रैखिक समीकरणों के निकाय (system of linear equations) पर विचार कीजिए।

$ax + 2y = \lambda$

$3x - 2y = \mu$

निम्नलिखित में से कौन सा (से) कथन सही है (हैं)? **[JEE-Adv.-2016]**

(A) यदि $a = -3$, तब λ और μ के सभी मानों के लिए निकाय के अनन्त (infinitely many) हल हैं

(B) यदि $a \neq -3$, तब λ और μ के सभी मानों के लिए निकाय का अद्वितीय (unique) हल है

(C) यदि $\lambda + \mu = 0$, तब $a = -3$ के लिए निकाय के अनन्त हल हैं

(D) यदि $\lambda + \mu \neq 0$, तब $a = -3$ के लिए निकाय का कोई हल नहीं है

Ans. BCD

Sol. System has unique solution for $\frac{a}{3} \neq \frac{2}{-2}$

system has infinitely many solution for $\frac{a}{3} = \frac{2}{-2} = \frac{\lambda}{\mu}$

and no solution for $\frac{a}{3} = \frac{2}{-2} \neq \frac{\lambda}{\mu}$

6. Let p, q, r be nonzero real numbers that are, respectively, the 10th, 100th and 1000th terms of a harmonic progression. Consider the system of linear equations **[JEE(Adv.)2022]**

$$x + y + z = 1$$

$$10x + 100y + 1000z = 0$$

$$qr\,x + pr\,y + pq\,z = 0$$

List - I	**List - II**
(I) If $\frac{q}{r} = 10$, then the system of linear equations has	(P) $x = 0, y = \frac{10}{9}, z = -\frac{1}{9}$ as a solution
(II) If $\frac{p}{r} \neq 100$, then the system of linear equations has	(Q) $x = \frac{10}{9}, y = -\frac{1}{9}, z = 0$ as a solution
(III) If $\frac{p}{q} \neq 10$, then the system of linear equations has	(R) infinitely many solutions
(IV) If $\frac{p}{q} = 10$, then the system of linear equations has	(S) no solution
	(T) at least one solution

The correct option is:

माना कि p, q, एवं r शून्येतर वास्तविक संख्यायें है जो एक हरात्मक श्रेढ़ी के क्रमशः 10वाँ, 100वाँ एवं 1000वाँ पद है। रैखिक समीकरणों के निकाय **[JEE Adv.-2022]**

$$x + y + z = 1$$

$$10x + 100y + 1000z = 0$$

$$qr\,x + pr\,y + pq\,z = 0$$

पर विचार कीजिए।

सूची - I

(I) यदि $\frac{q}{r} = 10$ है, तब रैखिक समीकरणों के निकाय का

(II) यदि $\frac{p}{r} \neq 100$ है, तब रैखिक समीकरणों के निकाय का

(III) यदि $\frac{p}{q} \neq 10$ है, तब रैखिक समीकरणों के निकाय का

(IV) यदि $\frac{p}{q} = 10$ है, तब रैखिक समीकरणों के निकाय का

सूची - II

(P) हल $x = 0, y = \frac{10}{9}, z = -\frac{1}{9}$ है

(Q) हल $x = \frac{10}{9}, y = -\frac{1}{9}, z = 0$ है

(R) अनंत हल है

(S) कोई हल नहीं है

(T) कम से कम एक हल है

सही विकल्प है :

(A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (R); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T)

(B) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (R)

(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

(D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (T)

Ans. (B)

Sol. $x + y + z = 1$ _____(1)

$10x + 100y + 1000z = 0$ _____(2)

$qrx + pry + pqz = 0$ _____(3)

Equation (3) can be re-written as

$$\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 0 \qquad (\because p, q, r \neq 0)$$

Let $p = \frac{1}{a+9d}, q = \frac{1}{a+99d}, r = \frac{1}{a+999d}$

Now, equation (3) is

$(a + 9d)x + (a + 99d)y + (a + 999d)z = 0$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 10 & 100 & 1000 \\ a+9d & a+99d & a+999d \end{vmatrix} = 0$$

$$\Delta_x = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 100 & 1000 \\ 0 & a+99d & a+999d \end{vmatrix} = 900(d-a)$$

$$\Delta_y = \begin{vmatrix} 1 & 1 & 1 \\ 10 & 0 & 1000 \\ a+9d & 0 & a+999d \end{vmatrix} = 990(a-d)$$

$$\Delta_z = \begin{vmatrix} 1 & 1 & 1 \\ 10 & 100 & 0 \\ a+9d & a+99d & 0 \end{vmatrix} = 90(d-a)$$

Option I: If $\frac{q}{r} = 10 \Rightarrow a = d$

$\Delta = \Delta_x = \Delta_y = \Delta_z = 0$

And eq. (1) and eq. (2) represents non-parallel planes and eq. (2) and eq. (3) represents same plane
$\Rightarrow$ Infinitely many solutions
I $\rightarrow$ P, Q, R, T

Option II : $\frac{p}{r} \neq 100 \Rightarrow a \neq d$

$\Delta = 0, \Delta_x, \Delta_y, \Delta_z \neq 0$

No solution
II $\rightarrow$ S

Option III : $\frac{p}{q} \neq 10 \Rightarrow a \neq d$

No solution
III $\rightarrow$ S

Option IV : If $\frac{p}{q} = 10 \Rightarrow a = d$

Infinitely many solution
IV $\rightarrow$ P, Q, R, T

7. Let α, β and γ be real numbers. Consider the following system of linear equations

$x + 2y + z = 7$

$x + \alpha z = 11$

$2x - 3y + \beta z = \gamma$

Match each entry in **List-I** to the correct entries in **List-II**.

List-I	**List-II**
(P) If $\beta = \frac{1}{2}(7\alpha - 3)$ and $\gamma = 28$, then the system has	(1)a unique solution
(Q) If $\beta = \frac{1}{2}(7\alpha - 3)$ and $\gamma \neq 28$, then the system has	(2)no solution
(R) If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where $\alpha = 1$ and $\gamma \neq 28$, then the system has	(3) infinitely many solutions
(S) If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where $\alpha = 1$ and $\gamma = 28$, then the system has	(4) $x = 11$, $y = -2$ and $z = 0$ as a solution
	(5) $x = -15$, $y = 4$ and $z = 0$ as a solution

The correct option is : **[JEE(Adv.)2023]**

माना कि α, β एवं γ वास्तविक संख्याएं (real numbers) हैं। निम्न रैखिक समीकरण निकाय (system of linear equation) पर विचार कीजिए।

$x + 2y + z = 7$

$x + \alpha z = 11$

$2x - 3y + \beta z = \gamma$

List-I की प्रत्येक प्रविष्ट (entry) का **List-II** की सही प्रविष्टियों (entries) से मिलान कीजिए।

List-I	**List-II**
(P) यदि $\beta = \frac{1}{2}(7\alpha - 3)$ एवं $\gamma = 28$, तब निकाय का(के)	(1) एक अद्वितीय हल (unique solution) है
(Q) यदि $\beta = \frac{1}{2}(7\alpha - 3)$ एवं $\gamma \neq 28$, तब निकाय का(के)	(2) कोई हल नहीं है
(R) यदि $\beta \neq \frac{1}{2}(7\alpha - 3)$ जहाँ $\alpha = 1$ एवं $\gamma \neq 28$, तब निकाय का(के)	(3) अनंत हल हैं
(S) यदि $\beta \neq \frac{1}{2}(7\alpha - 3)$ जहाँ $\alpha = 1$ एवं $\gamma = 28$, तब निकाय का(के)	(4) $x = 11$, $y = -2$ एवं $z = 0$ एक हल है
	(5) $x = -15$, $y = 4$ एवं $z = 0$ एक हल है

सही विकल्प है : **[JEE(Adv.)2023]**

(A) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (1); (S) $\rightarrow$ (4)

(B) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (5); (S) $\rightarrow$ (4)

(C) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (5)

(D) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (1); (S) $\rightarrow$ (3)

Ans. (A)

Sol. $D = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 0 & \alpha \\ 2 & -3 & \beta \end{vmatrix}$

$= 1\,[3\alpha] - 2\,[\beta - 2\alpha] + [-3]$

$= 3\alpha - 2\beta + 4\alpha - 3$

$= 7\alpha - 2\beta - 3$

If $D = 0$

$7\alpha - 2\beta - 3 = 0$

$\beta = \dfrac{7\alpha - 3}{2}$

$D_x = \begin{vmatrix} 7 & 2 & 1 \\ 11 & 0 & \alpha \\ \gamma & -3 & \beta \end{vmatrix}$

if $\gamma = 28$ & $\beta = \dfrac{7\alpha - 3}{2}$

then $D_x = \begin{vmatrix} 7 & 2 & 1 \\ 11 & 0 & \alpha \\ 28 & -3 & \dfrac{7\alpha - 3}{2} \end{vmatrix}$

$7[3\alpha] - 2\left[11\dfrac{(7\alpha - 3)}{2} - 28\alpha\right] + [-33]$

$21\alpha - [11\,(7\alpha - 3) - 56\alpha] - 33$

$21\alpha - (77\alpha - 33 - 56\alpha) - 33$

$= 0$

$D_y = \begin{vmatrix} 1 & 7 & 1 \\ 1 & 11 & \alpha \\ 2 & \gamma & \beta \end{vmatrix}$

Again if $\gamma = 28$ & $\beta = \dfrac{7\alpha - 3}{2}$

$D_y = \begin{vmatrix} 1 & 7 & 1 \\ 1 & 11 & \alpha \\ 2 & 28 & \dfrac{7\alpha - 3}{2} \end{vmatrix}$

$R_1 \to R_1 - R_2$

$$= \begin{vmatrix} 0 & -4 & 1-\alpha \\ 1 & 11 & \alpha \\ 2 & 28 & \dfrac{7\alpha-3}{2} \end{vmatrix}$$

$$4\left[\frac{7\alpha-3}{2}-2\alpha\right]+(1-\alpha)(28-22)$$

$2(7\alpha-3)-8\alpha+6(1-\alpha)$

$14\alpha-6-8\alpha+6-6\alpha$

$=0$

$$D_z = \begin{vmatrix} 1 & 2 & 7 \\ 1 & 0 & 11 \\ 2 & -3 & \gamma \end{vmatrix}$$

$1(33)-2(\gamma-22)+7(-3)$

$33+44-21-2\gamma$

$\Rightarrow 56-2\gamma$

If $\gamma = 28$

$D_z = 0$

(P) If $\beta = \frac{1}{2}(7\alpha-3)$ & $\alpha = 28$

then D_x, D_y, D_z are all zero

$\therefore$ Infinite many solution

$P \rightarrow (3)$

(Q) If $\beta = \frac{1}{2}(7\alpha-3)$ & $\gamma \neq 28$

It can be easily observed that $D_z \neq 0$

$\therefore D = 0$ & $D_z \neq 0$

$\Rightarrow$ No solution

$Q \rightarrow 2$

(R) If $\beta \neq \frac{1}{2}(7\alpha-3)$, i.e. $D \neq 0$

It will be the case of unique solution but here additional info of $\alpha = 1$ & $\gamma \neq 28$ is provided

(S) If $\alpha = 1$ then

$x+2y+z=7$

$x+z=11$

$2y=7-11$

$y=-2$ will always be the solution.

But $\gamma \neq 28$ then $z \neq 0$

So, $R \rightarrow (1)$

Again

If $\alpha = 1; \gamma = 28$

then $y = -2$; $z = 0$
& $z + 2(-2) = 7$
$x = 7 + 4$
$x = 11$
So we get the solution $S \rightarrow (4)$